Solution

Impulse Response Descriptions

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1. The impulse response of a system is the output when the input is a(n) _____ signal.

a) step

b) impulse

c) exponential

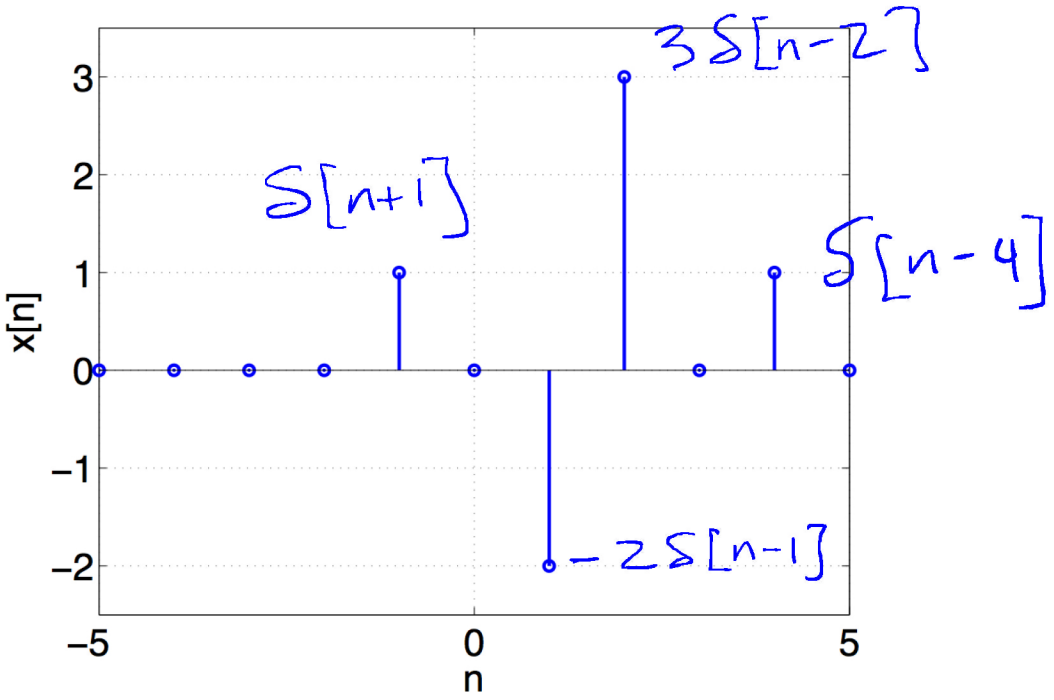
d) sinusoidal

2. The impulse response allows you to determine the output for any input if the system is linear and time invariant.

a) True

b) False

Problems 3-7 consider the discrete-time signal $x[n]$ is depicted in the figure below.



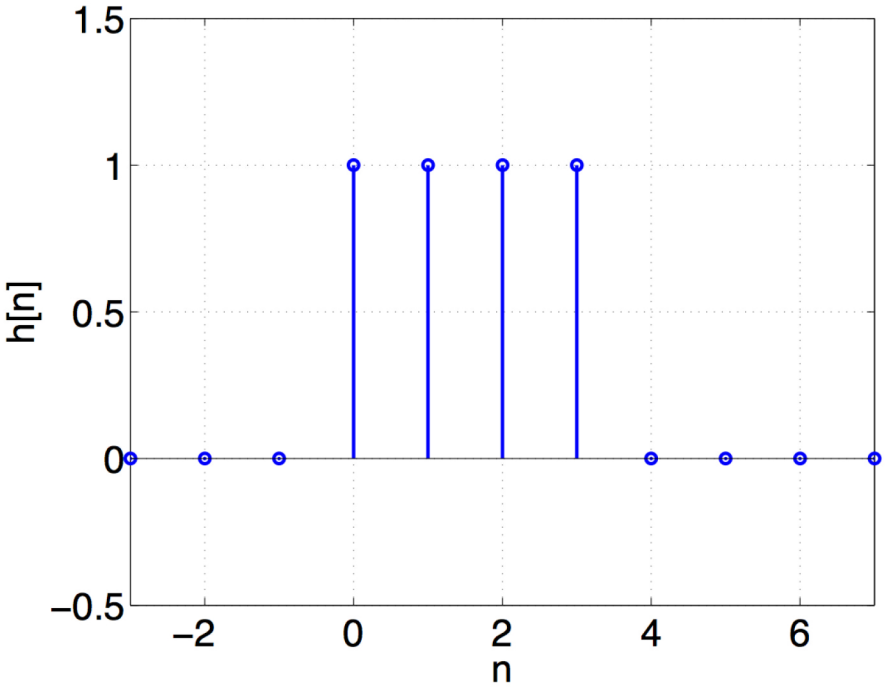
3. Express this signal as a weighted sum of time-shifted impulses.

$$x[n] = \delta[n+1] - 2\delta[n-1] + 3\delta[n-2] + \delta[n-4]$$

Problems 4-7 apply $x[n]$ to a LTI system with impulse response

$$h[n] = \begin{cases} 1 & n = 0, 1, 2, 3 \\ 0 & notherwise \end{cases}$$

as shown in the figure below.



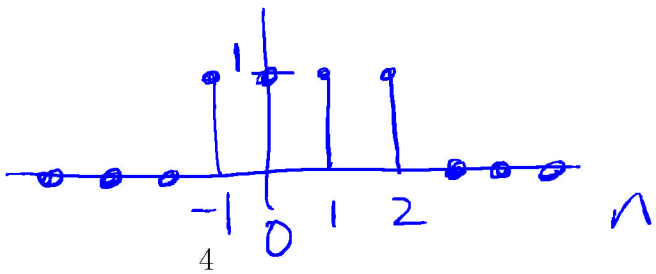
4. Evaluate the output $y[n]$ of this system if the input is the component of $x[n]$ associated with $\delta[n+1]$

by time invariance

$\delta[n-k] \rightarrow$ [h[n]] $\rightarrow h[n-k]$

LTI

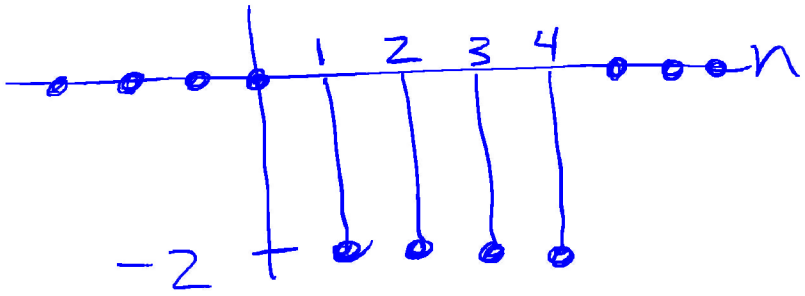
So $\delta[n+1] \rightarrow h[n+1]$



5. Evaluate the output $y[n]$ of this system if the input is the component of $x[n]$ associated with $\delta[n-1]$

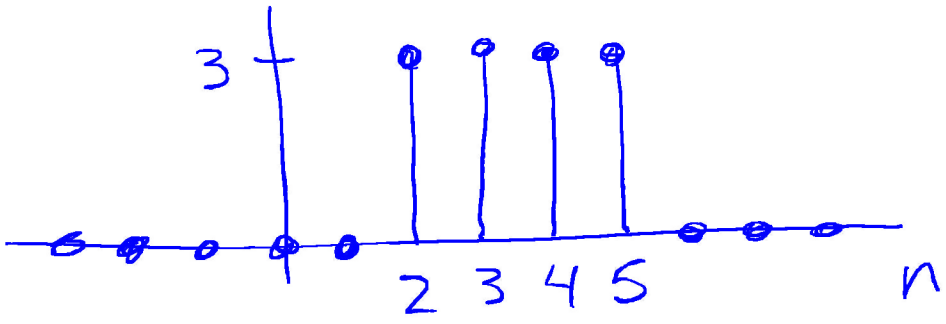
$\delta[n-1] \rightarrow h[n-1]$

by linearity $-2\delta[n-1] \rightarrow -2h[n-1]$



6. Evaluate the output $y[n]$ of this system if the input is the component of $x[n]$ associated with $\delta[n-2]$

$3\delta[n-2] \rightarrow 3h[n-2]$

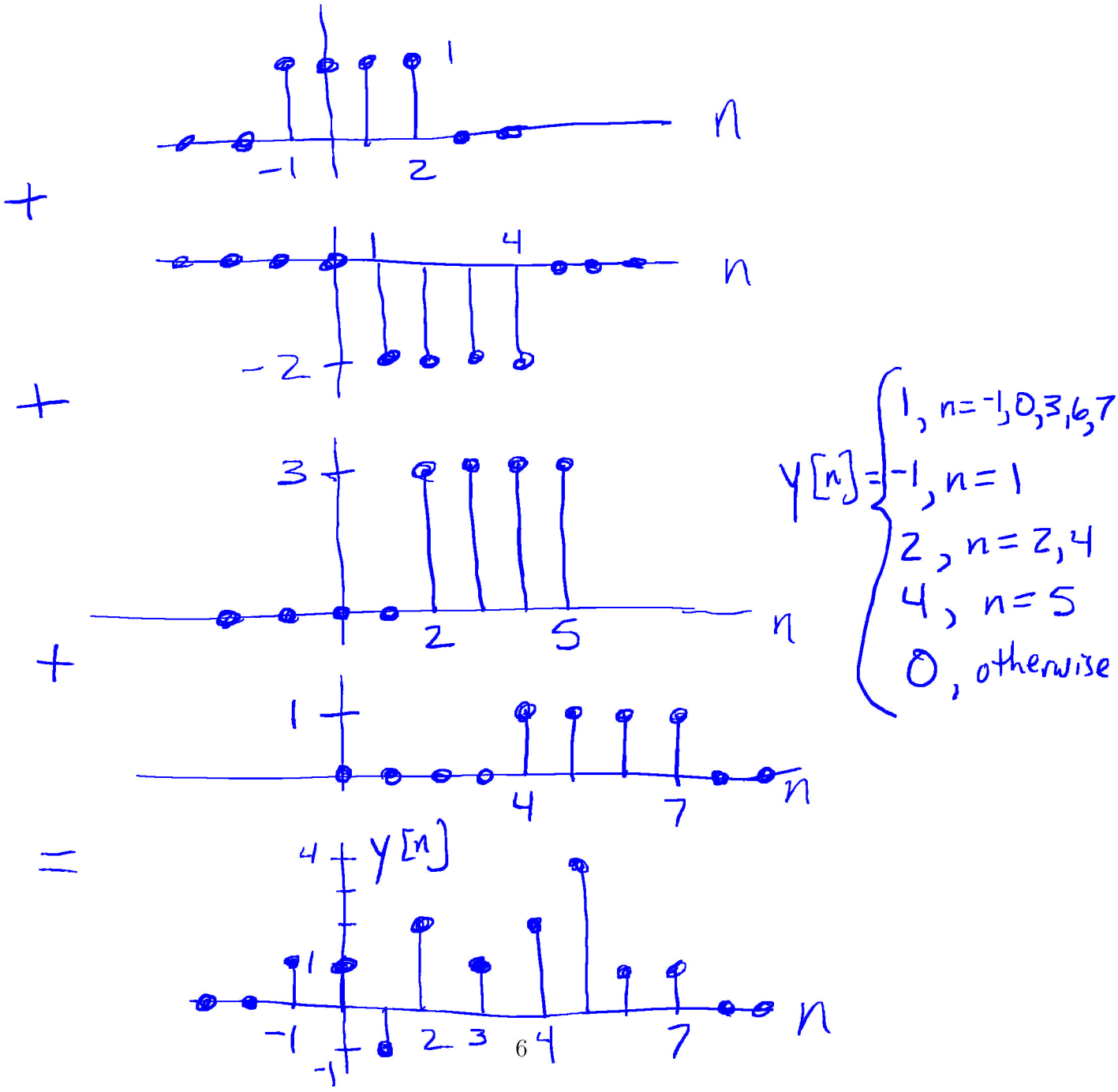


7. Evaluate the output $y[n]$ of this system if the input is $x[n]$.

$$x[n] = \delta[n+1] - 2\delta[n-1] + 3\delta[n-2] + \delta[n-4]$$

by linearity and time invariance

$$y[n] = h[n+1] - 2h[n-1] + 3h[n-2] + h[n-4]$$



$$y[n] = \begin{cases} 1, & n = -1, 0, 3, 6, 7 \\ -1, & n = 1 \\ 2, & n = 2, 4 \\ 4, & n = 5 \\ 0, & \text{otherwise} \end{cases}$$